Diary sentiment analysis:

Overview:

- The **Diary Sentiment Analysis** project allows users to write daily diary entries or short notes and automatically predicts whether the emotion behind the text is **positive or negative**. The system uses Natural Language Processing (NLP) and machine learning to understand the emotional tone of the user's writing. After every diary entry, the system also gives a **visual representation** of the user's emotional pattern by showing the percentage of positive entries recorded so far.

How to create it:

- Take data from Kaggle "twitter sentiment" which have sentence and a column of positive or negative sentence.
- Use pandas to load the data
- Handle the null value with fillna (method='ffill)
- Then make clean function remove punchuation.
- Divid the data using sklearn train,test,split
- Then x_train and x_test data converted to numeric from with the help of countvector()
- Use naïve bayes algorithm based model which is multinomialNB.
- Then fit in model x_train with y_train
- And then do testing with x_test
- Accuracy score(y_prep,y_test) is 0.73
- Then take diary entry from user store in csv file then perform this model
- Then count sentiment in data and then show the percentage of it .
- And also show graphically percentage distribution of sentiment.

How It Works (High-Level Flow)

1. **User writes a diary entry**
   - Put these data in csv file
2. **Text is cleaned** (removing special characters, converting to lowercase)
   - Clean the data, removing punctuation
3. **Model processes the text** using NLP techniques(MultinomialNB)
4. **Sentiment is predicted**
5. **Result is saved** in the database

   show percentage of sentiment
6. **Graph updates** to show overall emotional trend

Difficulties :

- Finding in dataset
- Finding model for this (then we understand for classification we have naïve bayes algo)
- Finding countvector()

🎯 Project Purpose

This system helps users:

- understand their daily emotional patterns
- reflect on their mental wellbeing
- maintain a long-term record of their mood
- track improvements or changes in sentiment

It can act as a simple emotional support tool by showing how the user's feelings evolve over weeks or months.

Library:

Pandas:

- Pandas is an open-source Python library used for data manipulation and analysis
- Provides time-series functionality
- We can easily handle missing data
- Faster data merging and joining
- Flexible reshaping of data sets
- Group by functionality
- Data from different file objects can be loaded
- Integrates with NumPy

Sklearn:

- Scikit-learn is an open-source Python library that simplifies the process of building machine learning models.
- Supports tasks like classification, regression, clustering and preprocessing
- Makes model building fast and reliable
- Provides ready-to-use tools for training and evaluation
- Reduces complexity by avoiding manual implementation of algo.